Forest cover type

Ironhack DSMLFT September 2026

Project 2 Machine Learning

23.10.2026

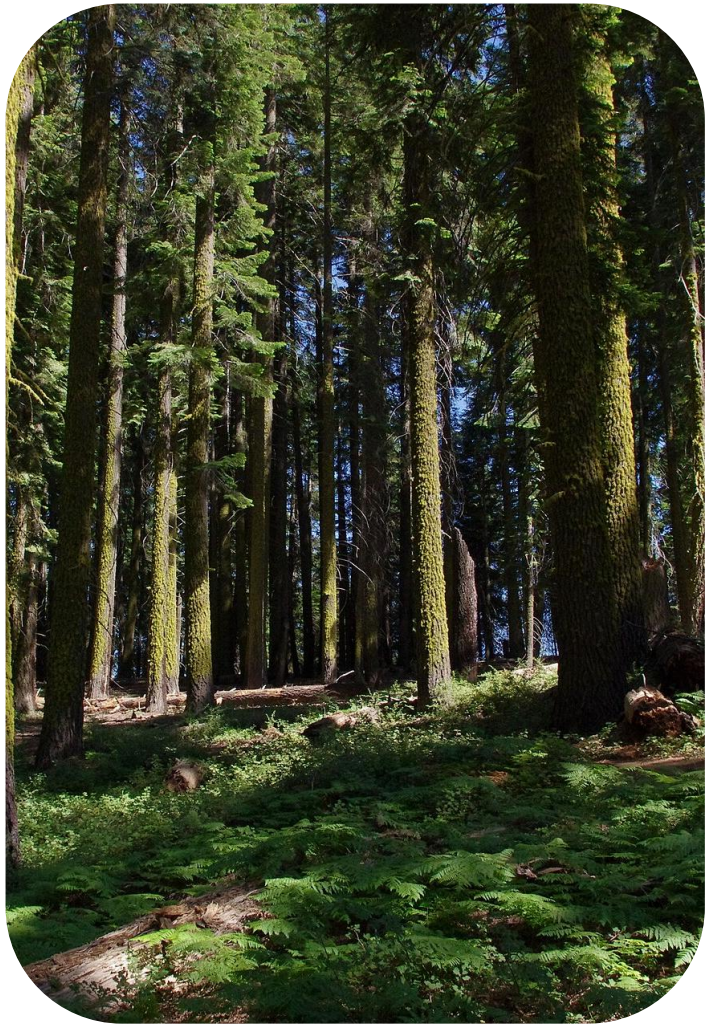
Zoé Doucet

What is a forest cover type?

The dominant tree species or vegetation type covering an area.



Aspen trees [1]



White fire forest [2]

Drivers

- Temperature
- Nutrients
- Water
- Sun exposure

Why do we need such a model?

The challenge

- Field survey cost time and money
- Not every forest is accessible

Solution

- Model that gives us the information on cover type
- Uses data we already have
- Needs to be:
  - fast
  - reliable

Why it matters

- Management
- Conservation and protection
- Risk assessment
- Decision making (logging, construction)

The Covertype dataset

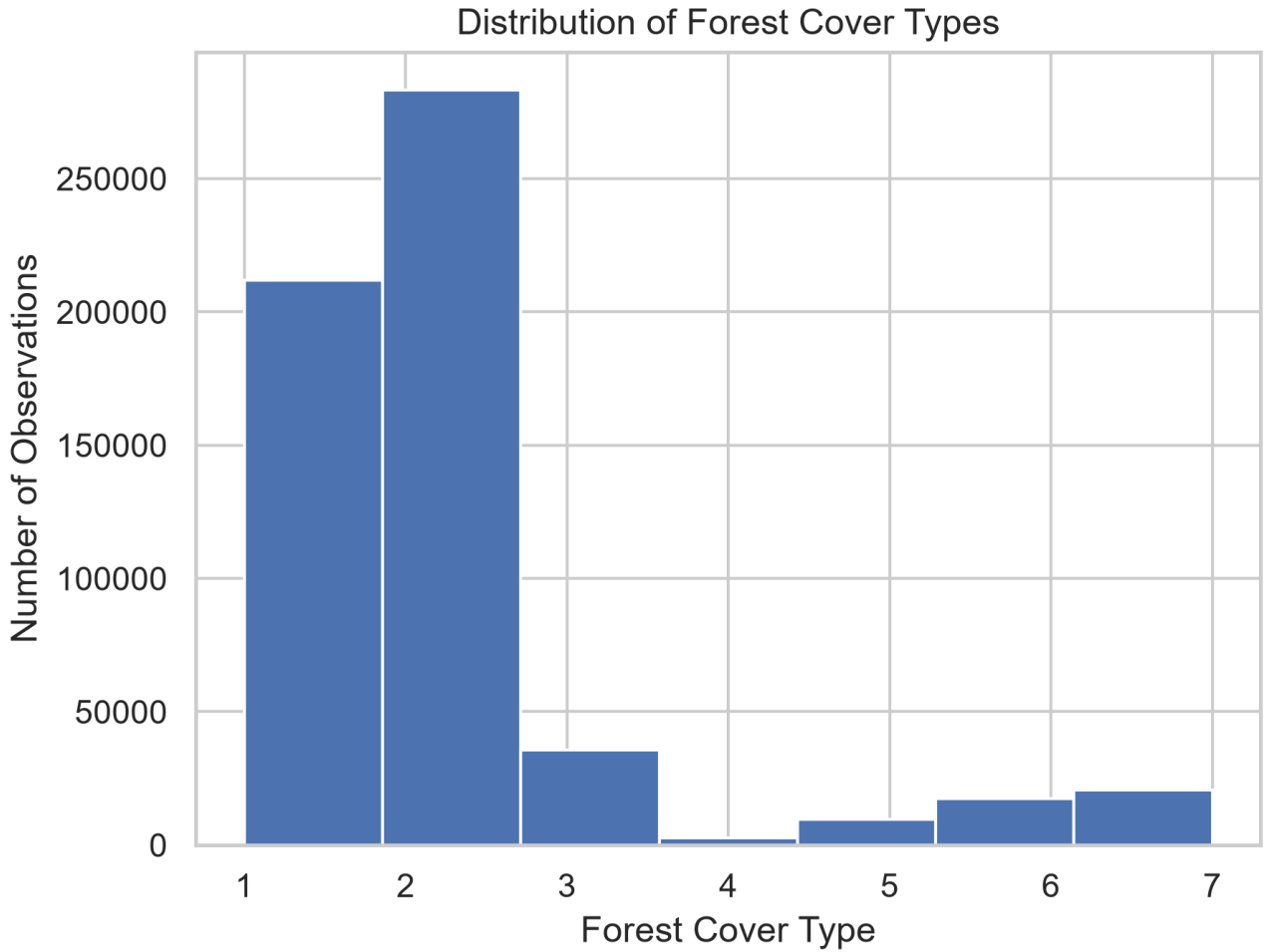
Created by **Jock Blackard**
Donated on **31/7/1998**

Downloaded from the UC Irvine Machine Learning Repository on the 15th of October 2026
DOI: https://doi.org/10.24432/C50K5N

Roosevelt National Forest Colorado, USA

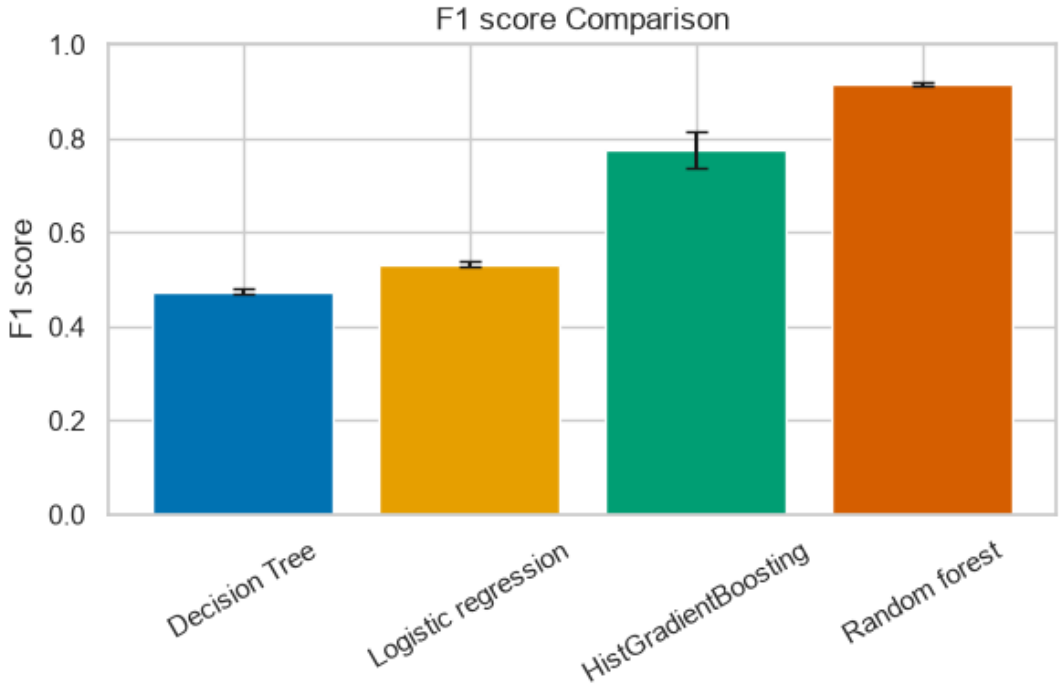
One row:

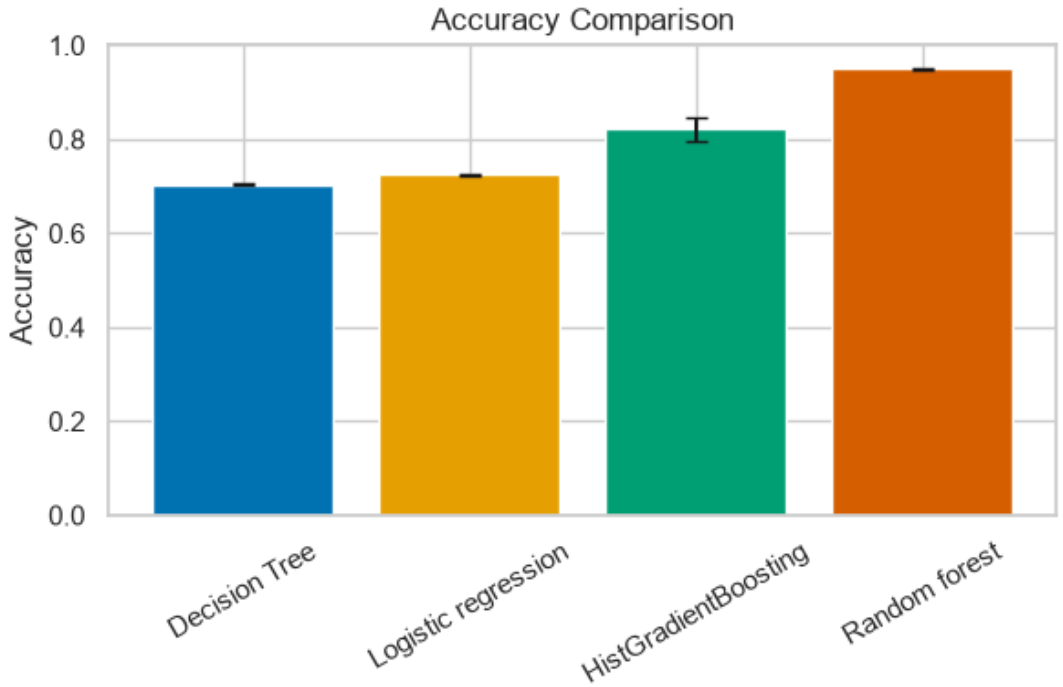
- 30 × 30 meter cell
- **features**: environmental and soil attributes
- **target**: one of seven forest cover types.



The right model

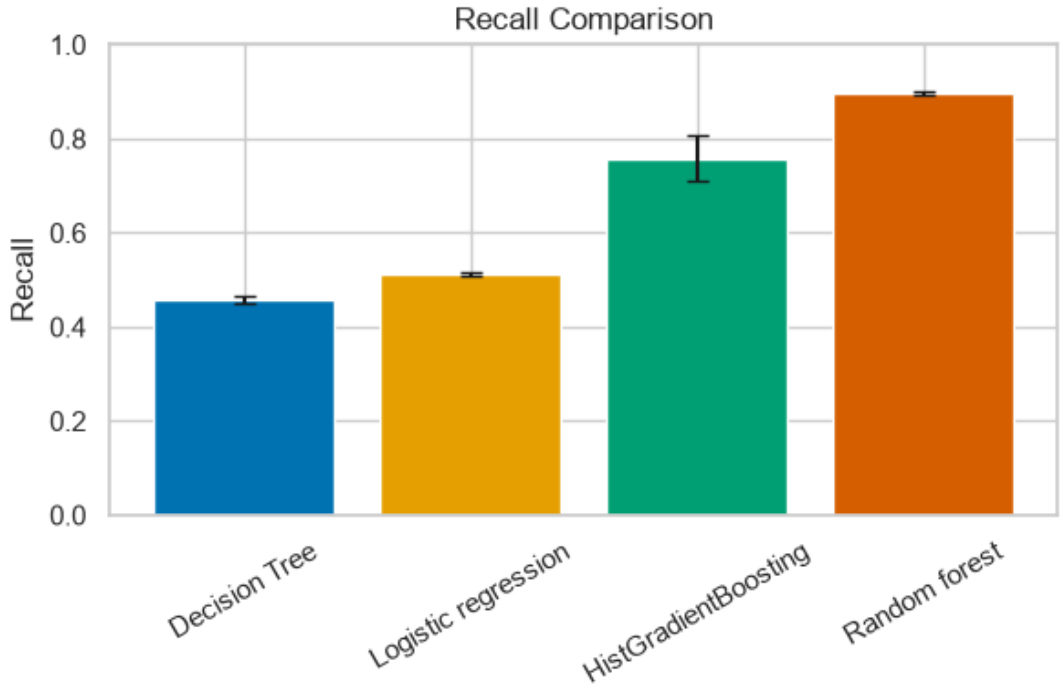
- **Random Forest:** best performance across all four metrics
  - **Accuracy**: **95.0** %,
  - F1 = 0.916
  - Precision = 0.940
  - Recall = 0.897
- Outperforms alternatives (accuracy):
  - HistGradientBoosting (82.2 %)
  - Logistic Regression (70.3 %)
  - Decision Tree (72.5 %)
- **High reliability**: minimal variation across cross-validation folds

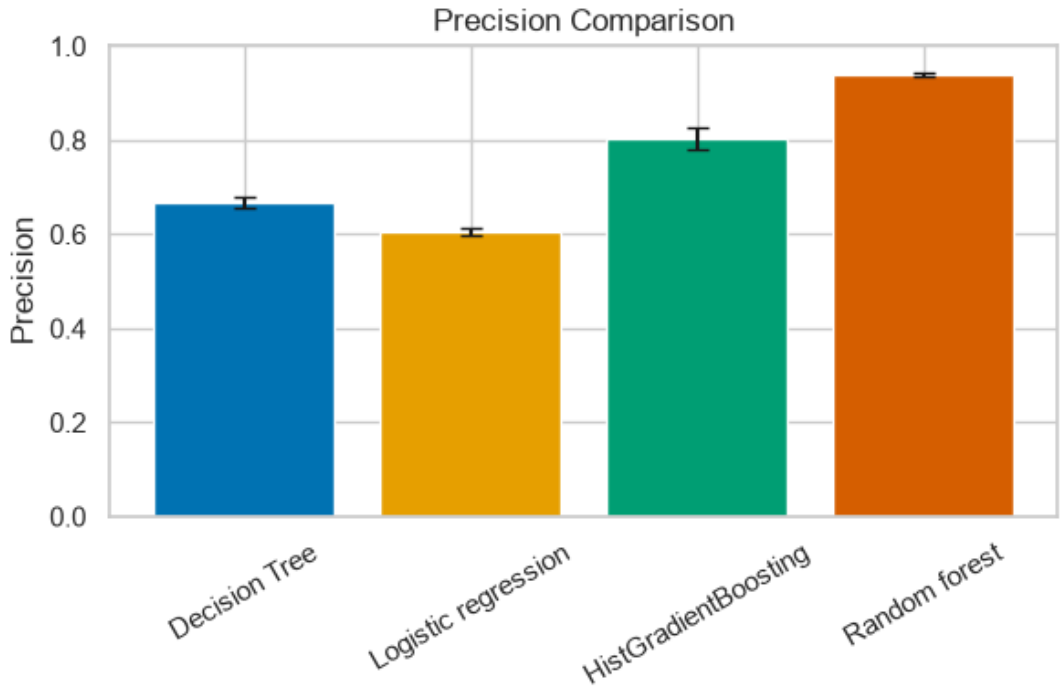




The right model

- **Random Forest:** best performance across all four metrics
  - Accuracy: 95.0 %,
  - F1 = 0.916
  - Precision = 0.940
  - **Recall = 0.897**
- Outperforms alternatives (recall):
  - HistGradientBoosting (0.759)
  - Logistic Regression (0.512)
  - Decision Tree (0.459)
- **High reliability**: minimal variation across cross-validation folds

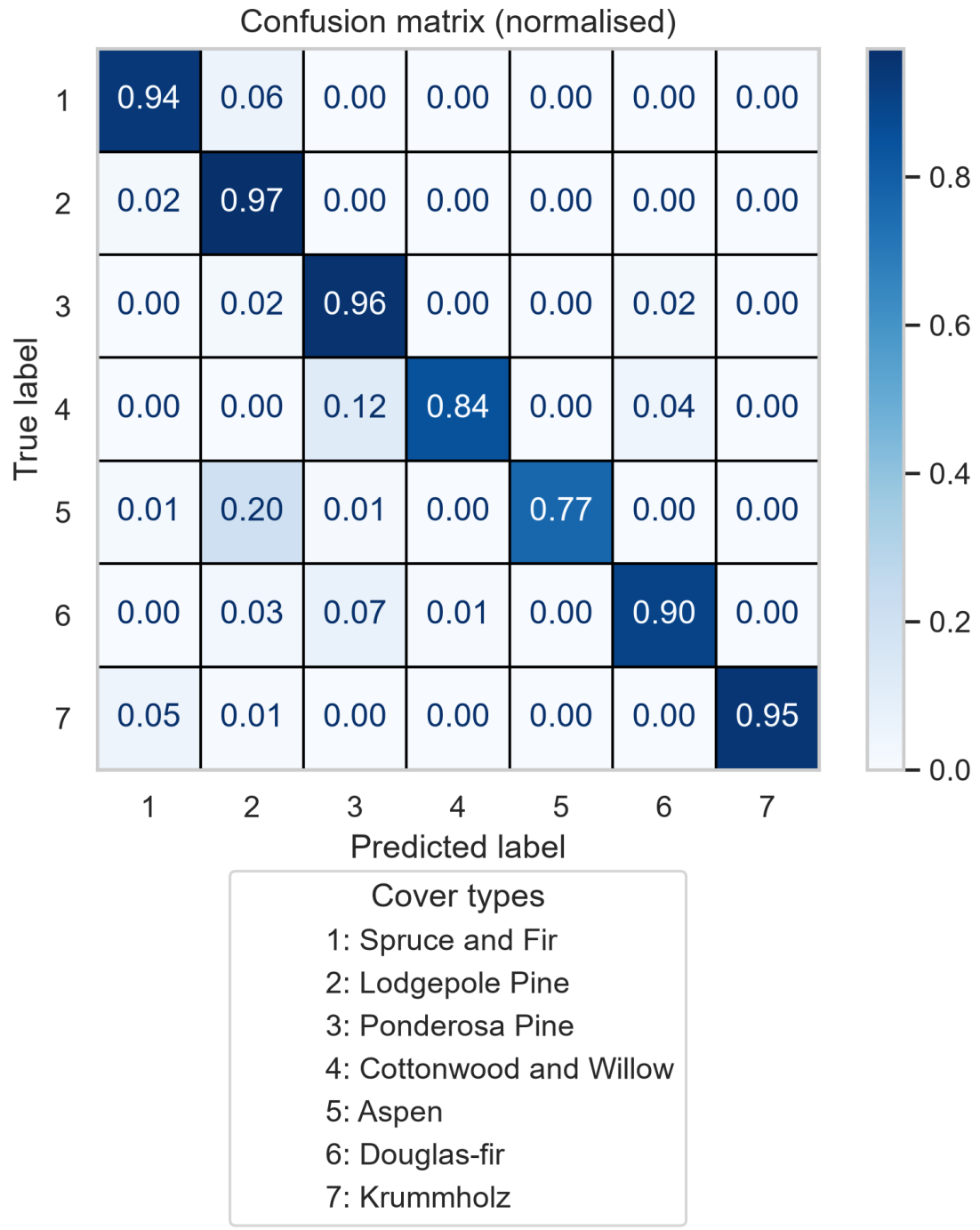




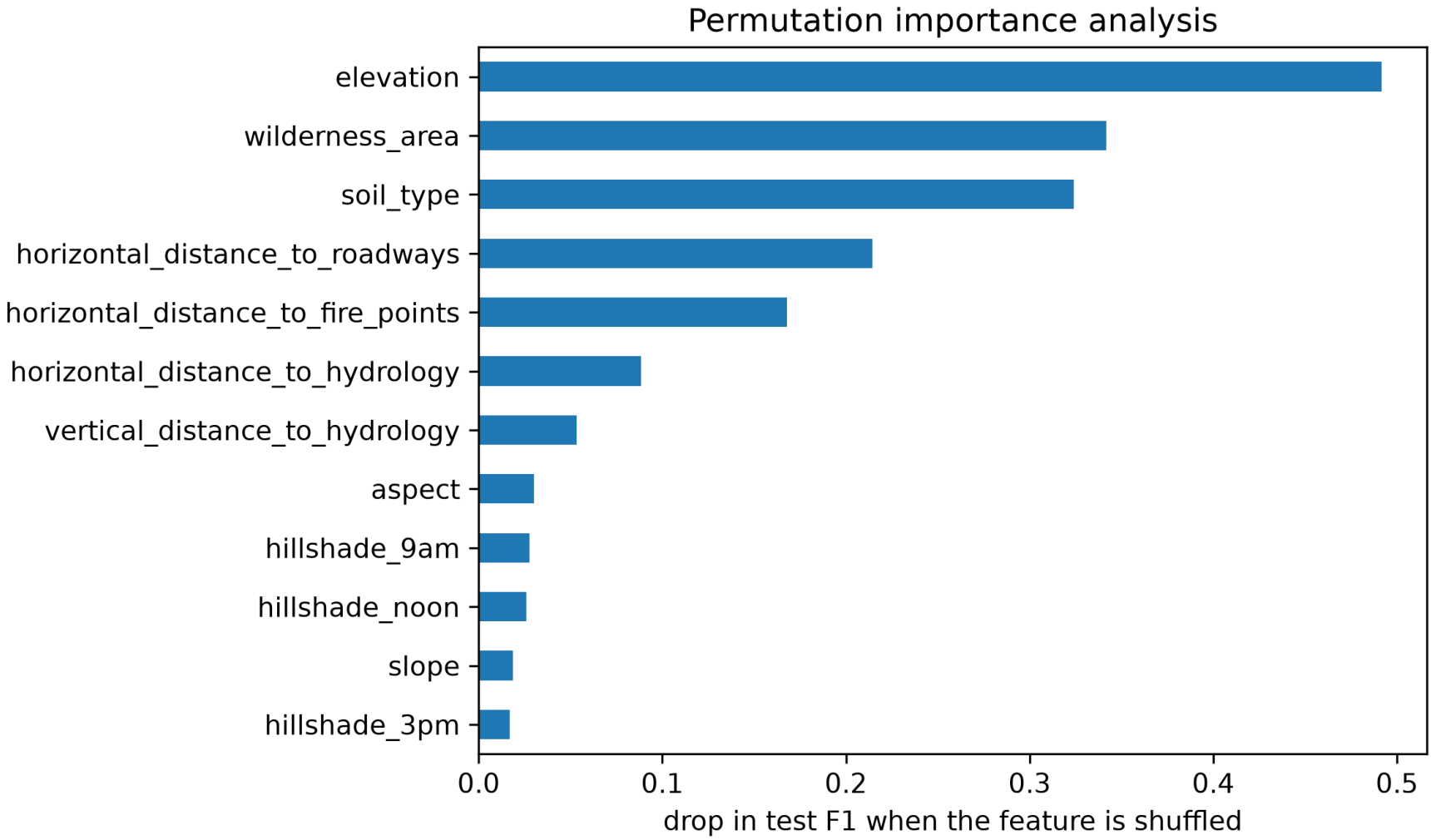
Confusion matrix

- **most cover types are correctly classified**
- Cover type **1 to 3, 6 and 7**:
  - correct classification 90 to 97 %
- Cover type **4**:
  - Correct classified 84 % of the time
  - misclassified as cover type 3 (12 %).
- Cover type **5**:
  - lowest correct classification (77 %)
  - misclassified as cover type 2 (20 %)

→ Changing the threshold does not improve the outcome



Most important features



- **Elevation** influences:
  - Temperature
  - UV radiation
  - Sun exposure
  - Wind
- **Wilderness area**:
  - geographic differences (micro climate, water)
- **Soil type** influences:
  - nutrient availability,
  - moisture retention

Demo using gradio



- Interactive demo of the trained forest cover type model
- User-friendly interface built with gradio
- No coding knowledge required to use the demo
- Input: sample number
- Predict forest cover type from the provided data

Conclusion

Practical use:

- **Predicting forest cover** in remote or inaccessible areas using cartographic data
- **Reliable** and **fast** for most forest cover types

Limitations:

- **Misclassification risk**: Aspen (class 5) often confused with Lodgepole Pine (class 2)
- **Real-world complexity**: unclear forest boundaries within 30 × 30 m cells

  → Not a replacement for field surveys when precise information is required

What to do next:

- **Improvements**
  - distinguishing Aspen (class 5) from Lodgepole Pine (class 2)
  - Is cartographic data alone sufficient?
- **Future research:**
  - investigate additional environmental variables
  - adjust classification thresholds for important cover types

Do you have any questions?

Fell free to ask me later!

Project completed during the

Data Science and Machine Leaning Bootcamp

at **Ironhack**

Project duration: 16/10/2026-23/10/2026

Lead teacher: Nicolas Mirabet

Technical assistant: Diana

Let‘s connect on LinkedIn!

Sources

Blackard, J. (1998). *Covertype*. UCI Machine Learning Repository. https://doi.org/10.24432/C50K5N

[1] https://commons.wikimedia.org/wiki/File:Stand_of_Aspen_trees_off_of_FR418_%28397149466 1%29.jpg

[2] https://commons.wikimedia.org/wiki/File:White_Fir_Forest_%287589428356%29.jpg